Name:_____

Exam Question Booklets
OCR Physics A

Booklet 5
11 Questions
103 marks

Things to remember:

✓

✓

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1

(a) Use the relation between force, mass and acceleration to express the newton in terms of the SI units of mass, length and time.

.....

.....[1]

(b) Fig. 3.1 shows a garden roller, of mass 80 kg, being pulled with a force of 250 N at an angle of 30° to the horizontal. The roller moves at a constant velocity.

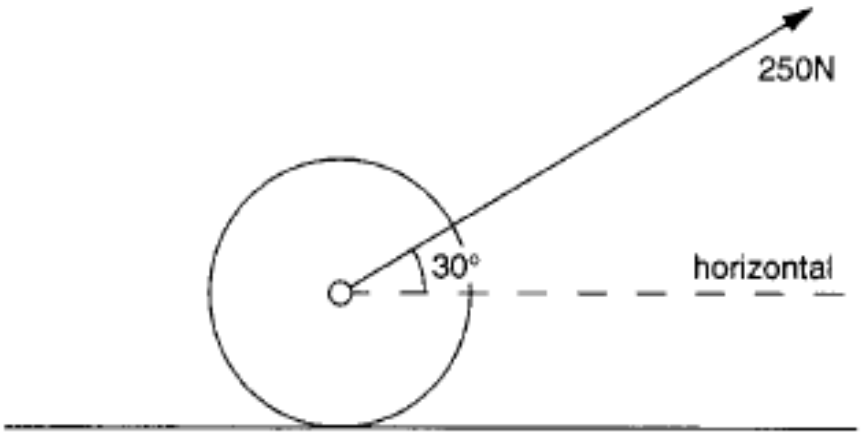

Fig. 3.1

(i) Calculate the weight of the roller.

weight = ... N [1]

(ii) Calculate the magnitude of the horizontal component of the force pulling the roller.

horizontal force = ... N [2]

(iii) Calculate the magnitude of the vertical force exerted by the roller on the ground.

vertical force = ... N [3]

(c) **(i)** Explain how the roller in **(b)** is able to travel at a constant velocity when being pulled by a force of 250 N.

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.....[2]

(ii) The roller is now pushed with a force of 250 N at 30° to the horizontal. State and explain the change that occurs to the force exerted on the ground by the roller. A numerical answer is not required.

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.....

.....[2]

2

(a) Define

(i) the *newton*

.....

..... [1]

(ii) *work done by a force.*

.....

..... [1]

(b) A car of mass m is travelling at a speed u. A constant braking force F is applied until the car comes to rest. The car travels a distance s after the force is applied.

(i) Write equations for

1 the work done by the force F

2 the relationship between the force F and acceleration a.

[1]

(ii) Use the equations in (i) to show that the work done bringing the car to rest equals the loss in kinetic energy of the car.

[3]

(iii) Hence describe the relationship between the braking distance and the velocity of a car.

.....

.....

..... [1]

(c) (i) A car of mass 1550 kg has a braking distance of 25.0 m when travelling at 13.3 m s^{-1}. Calculate the average braking force acting on the car.

force = ... N [2]

(ii) Deduce the braking distance when the car is travelling at 26.6 m s^{-1} and the same braking force is applied.

distance = ... m [1]

3

(a) **(i)** Define speed.....[1]

(ii) Distinguish between speed and velocity.

.....

.....[2]

(b) Use the equations given below, which represent uniformly accelerated motion in a straight line, to obtain an expression for v in terms of u, a and s only.

$v = u + at$

$s = (u + v)t/2$

[2]

(c) Fig. 1.1 shows a ball kicked from the top of a cliff with a horizontal velocity of $5.6\,\mathrm{m\,s^{-1}}$. Air resistance can be neglected.

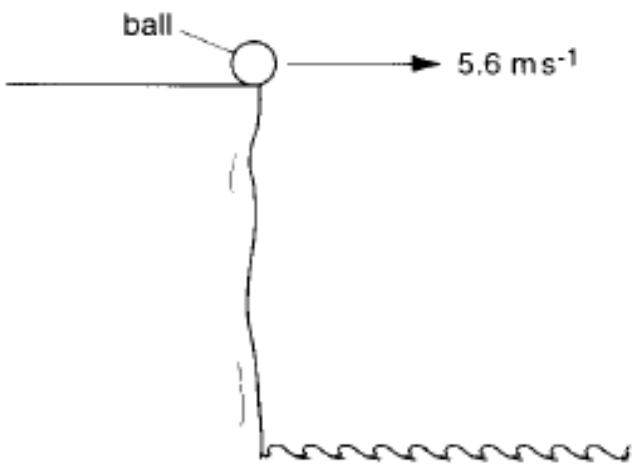

Fig. 1.1

(i) Show that after 0.90 s the vertical component of the velocity is $8.8\,\mathrm{m\,s^{-1}}$.

[2]

(ii) Use a vector triangle to determine the resultant velocity of the ball after 0.90 s.

resultant velocity: magnitude = ... $m\,s^{-1}$

angle to the horizontal = ... ° [4]

(iii) Calculate

1. the vertical distance the ball falls in 0.90 s,
2. the horizontal distance the ball travels in this time.

1. vertical distance = ... m
2. horizontal distance = ... m [3]

4 Fig. 3.1 shows the path of a golf ball from the time it ends contact with a golf club, point C, until it hits the ground at G. Assume that there is no air resistance.

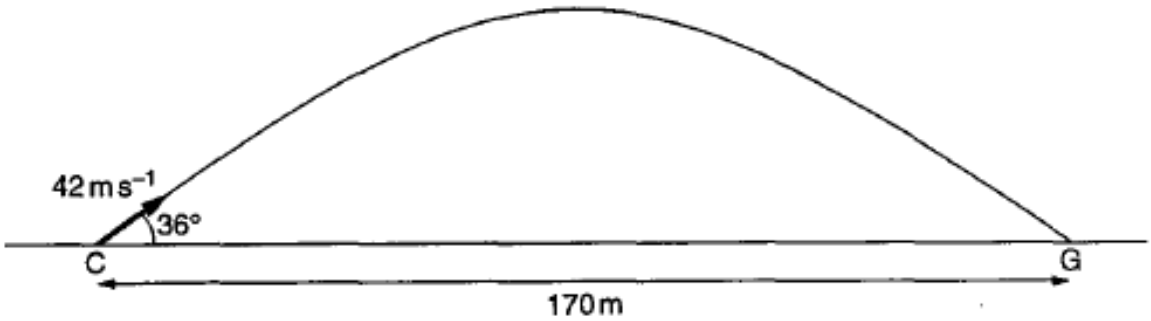

Fig. 3.1

The ball leaves the club with a velocity $42\,m\,s^{-1}$ at an angle of 36° to the horizontal.

(a) Show that the horizontal component of the velocity is $34\,m\,s^{-1}$

[1]

(b) The distance C to G is 170 m. Show that the time taken for the ball to travel from C to G is 5.0 s.

[1]

(c) Calculate

(i) the initial vertical component of the velocity

vertical velocity component = ... $m\,s^{-1}$ [2]

(ii) the maximum height reached.

maximum height = ... m [3]

(d) The ball has a mass of 50 g. Calculate the kinetic energy of the ball at maximum height.

kinetic energy = ... J [3]

(e) **(i)** On Fig. 3.1 sketch the path of the golf ball if air resistance is assumed **not** to be negligible. [2]

(ii) Explain the shape of your sketch.

.....

.....[1]

5

(a) Define

(i) power .....

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.....[1]

(ii) a joule .....

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.....[1]

(b) Fig. 4.1 shows part of a fairground ride with a carriage on rails.

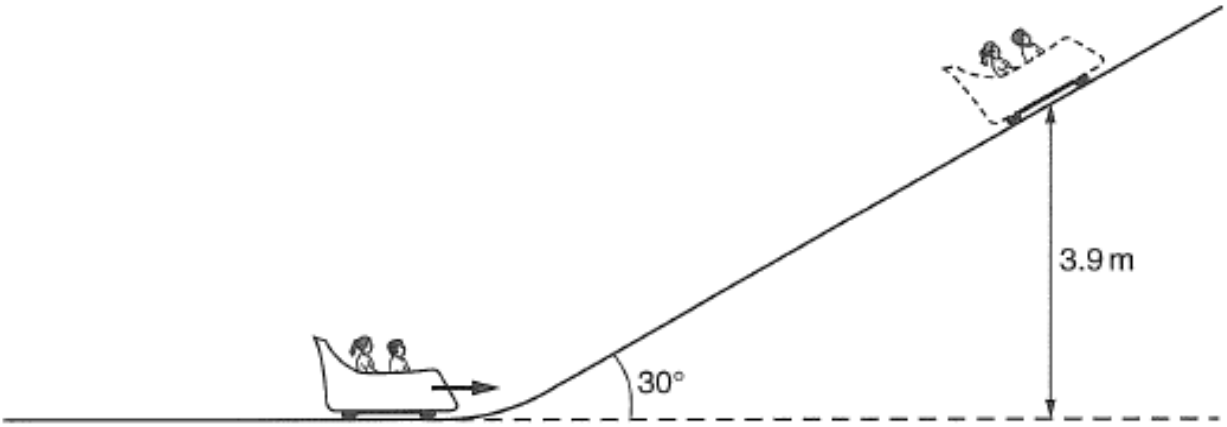

Fig. 4.1

The carriage of mass 500 kg is travelling towards a slope inclined at 30° to the horizontal. The carriage has a kinetic energy of 25 kJ at the bottom of the slope. The carriage comes to rest after travelling up the slope to a vertical height of 3.9 m.

(i) Show that the potential energy gained by the carriage is 19 kJ.

[2]

(ii) Calculate the work done against the resistive forces as the carriage moves up the slope.

work done = ... kJ [1]

(iii) Calculate the resistive force acting against the carriage as it moves up the slope.

resistive force = ... N [3]

6 Fig. 1.1 illustrates a conveyor belt for transporting young children up a snow-covered bank so that they can ski back down.

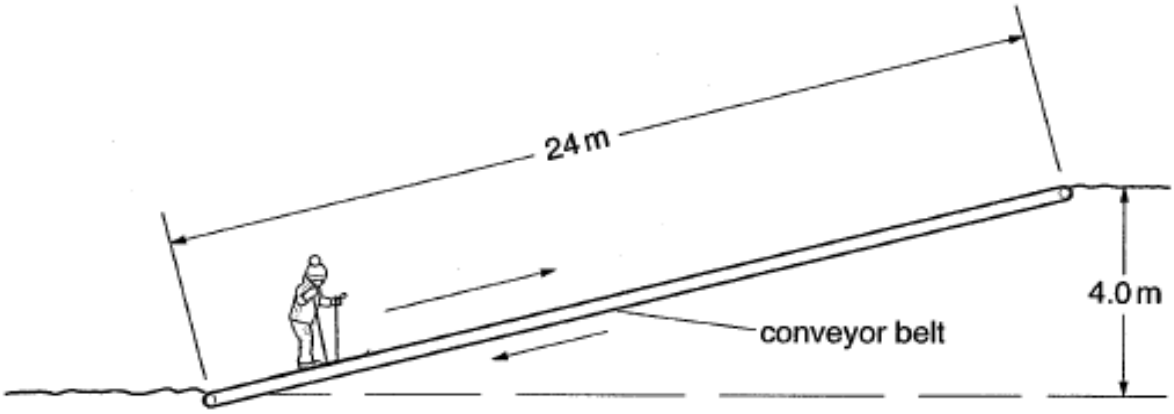

Fig. 1.1

A child of mass 20 kg travels up the conveyor belt at a constant speed. The distance travelled up the slope is 24 m and the time taken is 55 s. The vertical height climbed in this time is 4.0 m.

(a) For the child on the conveyor belt, calculate

(i) her speed

speed = ... m s^{-1} [2]

(ii) her kinetic energy

kinetic energy = ... J [2]

(iii) the increase in her potential energy for the complete journey up the slope.

potential energy = ... J [2]

(b) **(i)** The conveyor belt is designed to take a maximum of 15 children at any one time. Calculate the power needed to lift 15 children of average mass 20 kg through a height of 4.0 m in 55 s.

power = ... W [2]

(ii) The belt is driven by an electric motor. State **two** reasons why the motor needs a greater output power than that calculated in **(b)(i)**.

.....

.....

.....

.....[2]

7

The front of a lorry can be considered a flat vertical surface of area 8.0 m^2. The average pressure due to air resistance is 150 Pa when the lorry travels at a constant speed of 16 $m\,s^{-1}$. You may assume the air resistance force acts perpendicular to the front of the lorry. When the lorry travels at 16 $m\,s^{-1}$

(i) calculate the force due to the air resistance

force due to air resistance = ... N [2]

(ii) calculate the power dissipated in overcoming the air resistance.

power = ... W [3]

(iii) State and explain how the values of the quantities calculated above would change if the lorry increased its speed.

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.....[3]

A ball leaves point P at the top of a cliff with a horizontal velocity of $15\,m\,s^{-1}$, as shown in Fig. 2.1.

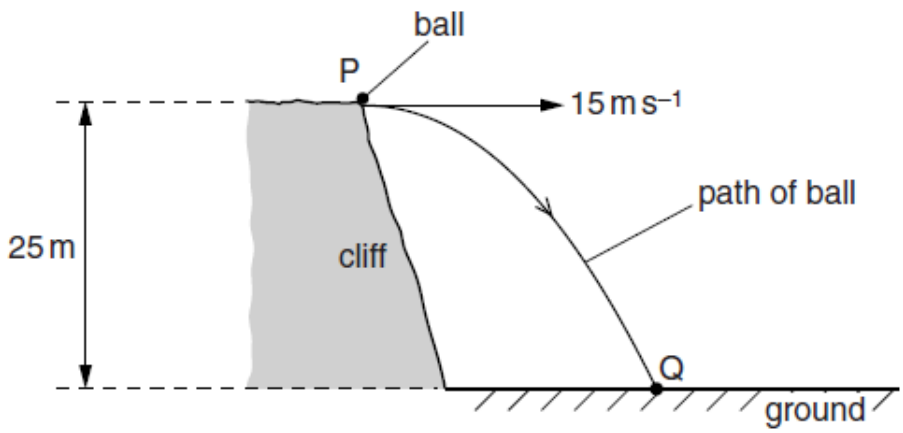

Fig. 2.1

The height of the cliff is 25 m. The ball hits the ground at point Q.
Air resistance is negligible.

(i) Calculate the vertical velocity of the ball just before it makes impact with the ground at Q.

vertical velocity = ... $m\,s^{-1}$ [2]

(ii) Show that the time taken for the ball to fall to the ground is 2.3 s.

[1]

(iii) Calculate the magnitude of the displacement of the ball at point Q from point P.

displacement = ... m [4]

(iv) Explain why the distance travelled by the ball is different from the magnitude of the displacement of the ball.

.....

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.....[2]

9

A box slides down a slope, as shown in Fig. 3.1.

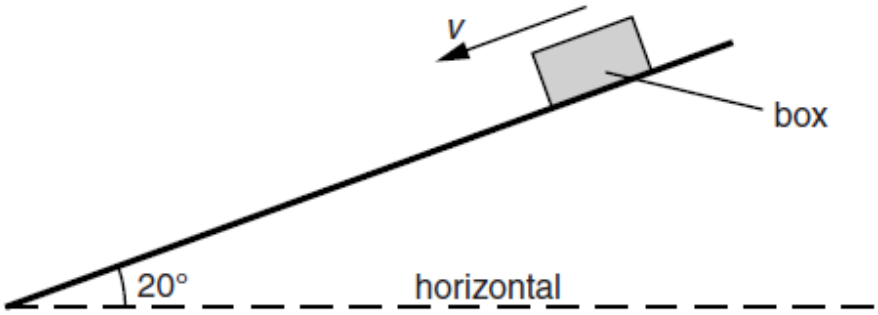

Fig. 3.1

The angle of the slope to the horizontal is 20°. The box has a mass of 65 kg. The total resistive force *R* acting on the box is constant as it slides down the slope.

(i) State the names and directions of the other two forces acting on the box.

1. .....

2. .....

[2]

(ii) The variation with time t of the velocity v of the box as it moves down the slope is shown in Fig. 3.2.

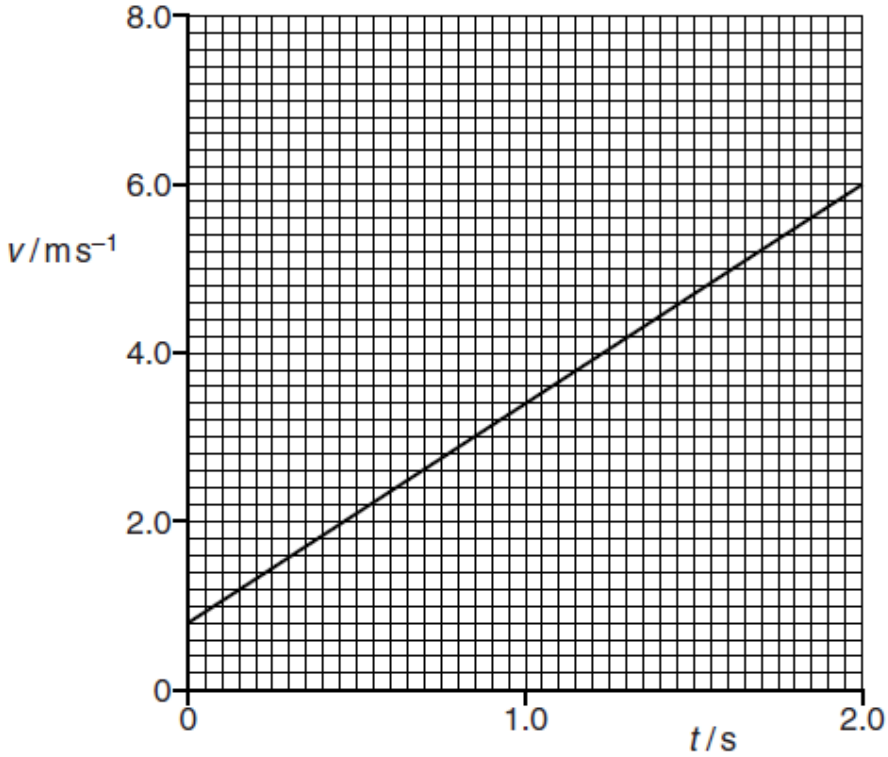

Fig. 3.2

1. Use data from Fig. 3.2 to show that the acceleration of the box is 2.6 $m\,s^{-2}$.

[2]

2. Calculate the resultant force on the box.

resultant force = ... N [1]

3. Determine the resistive force R on the box.

R = ... N [3]

A ball of mass 400 g is thrown with an initial velocity of $30.0\,m\,s^{-1}$ at an angle of 45.0° to the horizontal, as shown in Fig. 4.1.

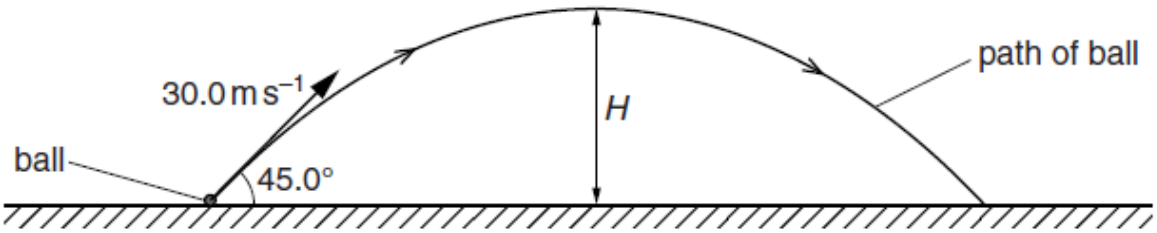

Fig. 4.1

Air resistance is negligible. The ball reaches a maximum height *H* after a time of 2.16 s.

(i) Calculate

1. the initial kinetic energy of the ball,

kinetic energy = ... J [3]

2. the maximum height *H* of the ball,

H = ... m [2]

3. the gravitational potential energy of the ball at height *H*.

potential energy = ... J [2]

(ii) **1.** Determine the kinetic energy of the ball at its maximum height.

kinetic energy = ... J [1]

2. Explain why the kinetic energy of the ball at maximum height is not zero.

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..... [1]

11

An object falls vertically from rest through air. State and explain the energy conversions that occur as the object falls.

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..... [3]